

Topic C3

Stoichiometry

Concept guide · Worked examples · Cambridge-style practice · Fully explained answers

What this guide teaches

A complete explanation of the syllabus concepts, chemical reasoning, worked applications, common misconceptions and original exam-style practice for this topic.

Syllabus map

C3.1	Formulae and equations
C3.2	Relative atomic and molecular masses
C3.3	The mole and Avogadro constant
C3.4	Reacting masses
C3.5	Limiting reactants and yield
C3.6	Concentration and gas volumes

How to study: retrieve definitions and equations, cover the worked examples, then complete the practice paper without using the notes.

1. Formulae and equations

A formula shows the simplest whole-number ratio of atoms or ions. Balanced equations conserve the number of each type of atom and therefore mass. State symbols are (s), (l), (g) and (aq). Ionic equations remove spectator ions and show only species that change.

Worked example

$\text{Mg} + 2\text{HCl} \rightarrow \text{MgCl}_2 + \text{H}_2$; coefficients balance atoms but subscripts in formulae must not be changed.

Examiner insight

Balance with coefficients only. Changing a subscript changes the substance.

2. Relative masses and the mole

Relative molecular/formula mass, M_r , is the sum of A_r values. One mole contains the Avogadro constant, 6.02×10^{23} , of specified particles. Molar mass has units g/mol.

$$n = m/M \quad N = nL$$

Worked example

9.0 g of water, M_r 18, is $9.0/18 = 0.50$ mol, containing 3.01×10^{23} molecules.

Examiner insight

Always state whether particles are atoms, molecules, ions or formula units.

3. Using equation ratios

Coefficients give mole ratios. Convert the known quantity to moles, apply the ratio, then convert to the required quantity. The limiting reactant is consumed first and fixes the maximum product amount.

known quantity $\rightarrow$ moles $\rightarrow$ mole ratio $\rightarrow$ required quantity

Worked example

$2\text{Mg} + \text{O}_2 \rightarrow 2\text{MgO}$. 0.30 mol Mg forms 0.30 mol MgO.

Examiner insight

Do not apply a mass ratio directly unless it has been calculated from molar masses.

4. Empirical formula and composition

An empirical formula is the simplest whole-number atom ratio. Convert each mass or percentage to moles, divide by the smallest and scale to whole numbers. Molecular formula is a whole-number multiple of the empirical formula.

percentage by mass = $\frac{\text{mass of element}}{\text{formula mass}} \times 100$

Worked example

A compound has 24 g C and 4 g H: mole ratio $24/12 : 4/1 = 2:4 = 1:2$, so empirical formula CH_2 .

Examiner insight

Use unrounded ratios until the final whole-number step.

5. Concentration, gas volume and yield

Concentration may be expressed in g/dm^3 or mol/dm^3 . At room temperature and pressure, one mole of gas occupies 24 dm^3 . Percentage yield compares actual with theoretical product; purity compares desired material with total sample.

$c = n/V$ gas moles = $V/24 \text{ dm}^3$ yield = $\frac{\text{actual}}{\text{theoretical}} \times 100$

Worked example

25.0 cm^3 of 0.200 mol/dm^3 solution contains $0.200 \times 0.0250 = 0.00500 \text{ mol}$.

Examiner insight

Convert cm^3 to dm^3 by dividing by 1000 before using molar concentration.

Exam technique

High-value habits

Use precise chemical vocabulary, balance equations, show every calculation step, include units and state symbols where required, distinguish observations from conclusions, and answer the command word directly.

Retrieval checklist

- Recall all definitions, tests and equations without notes.
- Explain one unfamiliar observation using particles, bonding or collision theory.
- Check charge, atoms, units and significant figures in every calculation.
- Complete one timed set and mark it only after finishing.

Practice Paper

Topic C3

stoichiometry**Instructions**

Answer all questions. Select one option for multiple choice. Show working in open spaces and write explanations on the lines. Individual parts carry 1-3 marks.

Candidate name:

Class / date:

Total marks: 50

Suggested time: 60 minutes.

Section A · Multiple choice

Choose the best answer for each question.

1. What is the M_r of CaCO_3 ? Ar: Ca 40, C 12, O 16.

[1]

- A 68
- B 84
- C 100
- D 116

2. How many moles are in 11 g of CO_2 ?

[1]

- A 0.25
- B 0.50
- C 2.0
- D 4.0

3. Which is a correct balanced equation?

[1]

- A $\text{H}_2 + \text{O}_2 \rightarrow \text{H}_2\text{O}$
- B $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$
- C $\text{H}_2 + 2\text{O}_2 \rightarrow \text{H}_2\text{O}$
- D $2\text{H}_2 + 2\text{O}_2 \rightarrow \text{H}_2\text{O}$

4. How many particles are in 2.0 mol?

[1]

- A 3.01×10^{23}
- B 6.02×10^{23}
- C 1.20×10^{24}
- D 2.00×10^{24}

5. At r.t.p., 0.50 mol gas occupies

[1]

- A 6 dm³
- B 12 dm³
- C 24 dm³
- D 48 dm³

6. 0.20 mol solute in 0.50 dm³ has concentration

[1]

- A 0.10
- B 0.40
- C 2.5
- D 4.0

7. What is the empirical formula for mole ratio C:H = 1:3?

[1]

- A CH
- B CH₂
- C CH₃
- D C₂H₆

8. The limiting reactant is the reactant that [1]
- A has the largest mass
 - B is used up first
 - C has the highest M_r
 - D is always a gas
9. Actual yield 8.0 g; theoretical yield 10.0 g. Percentage yield? [1]
- A 20%
 - B 80%
 - C 100%
 - D 125%
10. Which ionic equation represents acid neutralisation? [1]
- A $H^+ + OH^- \rightarrow H_2O$
 - B $Na^+ + Cl^- \rightarrow NaCl$
 - C $H_2 \rightarrow 2H^+$
 - D $O_2^- + H_2O \rightarrow O_2$

Section B · Structured questions

11 Calcium carbonate has formula CaCO_3 . Ar: Ca 40, C 12, O 16.

(a) Calculate its relative formula mass. [2]

(b) Calculate moles in 25.0 g. [2]

(c) Calculate the number of formula units. [2]

(d) Calculate the percentage by mass of calcium. [2]

[Total: 8]

12 $2\text{Al} + 3\text{Cl}_2 \rightarrow 2\text{AlCl}_3$. A student reacts 0.40 mol Al with excess chlorine.

(a) State the Al:AlCl₃ mole ratio. [1]

(b) Calculate moles of AlCl₃. [1]

(c) Calculate its mass. Mr(AlCl₃)=133.5. [3]

(d) Explain what “excess chlorine” means. [3]

[Total: 8]

13 A compound contains 40.0% carbon, 6.7% hydrogen and 53.3% oxygen.

(a) Calculate the mole ratio. Ar: C 12, H 1, O 16. [3]

(b) Deduce the empirical formula. [2]

(c) Its Mr is 180. Deduce the molecular formula. [2]

(d) Define empirical formula. [1]

[Total: 8]

14 25.0 cm³ of 0.200 mol/dm³ NaOH neutralises HCl.

(a) Convert the NaOH volume to dm³. [1]

(b) Calculate moles of NaOH. [2]

(c) Using $\text{HCl} + \text{NaOH} \rightarrow \text{NaCl} + \text{H}_2\text{O}$, find moles of HCl. [1]

(d) If HCl volume is 20.0 cm^3 , calculate its concentration. [3]

(e) State the ionic equation. [1]

[Total: 8]

15 A reaction should produce 12.0 g but gives 9.0 g of pure product.

(a) Calculate percentage yield. [2]

(b) State two reasons for a yield below 100%. [2]

(c) A 15.0 g impure sample contains 12.0 g product. Calculate purity. [2]

(d) At r.t.p., calculate the volume of 0.25 mol gas. [2]

[Total: 8]

Worked solutions

Remove or fold back this section while completing the practice paper.

Marking routine

Mark each part separately. Award method and chemistry marks where shown, then identify the first point at which an answer becomes incorrect.

Solutions · Multiple choice

1 C. $40 + 12 + 3 \times 16 = 100$.

2 A. M_r 44, so $11/44 = 0.25$ mol.

3 B. It conserves four H and two O atoms.

4 C. $2.0 \times 6.02 \times 10^{23}$.

5 B. $0.50 \times 24 = 12$ dm³.

6 B. $c = n/V = 0.20/0.50$.

7 C. The simplest whole-number ratio is 1:3.

8 B. Once consumed, no more product can form.

9 B. $8.0/10.0 \times 100 = 80\%$.

10 A. Hydrogen and hydroxide ions form water.

Solutions · Structured questions

11. 11(a) $40+12+48 = 100$ [2]. (b) $25.0/100 = 0.250$ mol [2]. (c) $0.250 \times 6.02 \times 10^{23} = 1.51 \times 10^{23}$ formula units [2]. (d) $40/100 \times 100 = 40\%$ [2].

12. 12(a) 2:2 or 1:1 [1]. (b) 0.40 mol [1]. (c) $0.40 \times 133.5 = 53.4$ g, with method and unit [3]. (d) More chlorine is present than required [1]; aluminium is limiting [1]; chlorine remains after reaction [1].

13. 13(a) C $40/12=3.33$; H 6.7; O $53.3/16=3.33$, ratio 1:2:1 [3]. (b) CH₂O [2]. (c) empirical mass 30; multiplier 6; C₆H₁₂O₆ [2]. (d) Simplest whole-number ratio of atoms [1].

14. 14(a) 0.0250 dm³ [1]. (b) $0.200 \times 0.0250 = 0.00500$ mol [2]. (c) 0.00500 mol [1]. (d) $0.00500/0.0200 = 0.250$ mol/dm³ [3]. (e) $H^+ + OH^- \rightarrow H_2O$ [1].

15. 15(a) $9.0/12.0 \times 100 = 75\%$ [2]. (b) Incomplete/reversible reaction, side reactions, loss during transfer/purification [any 2]. (c) $12.0/15.0 \times 100 = 80\%$ [2]. (d) $0.25 \times 24 = 6.0$ dm³ [2].

Self-assessment

Circle the first incorrect step and classify it as knowledge, equation or formula, mole ratio, unit conversion, observation, practical method or communication.

Evidence used: Cambridge IGCSE Co-ordinated Sciences 0654 syllabus 2025-2027 and June 2026 papers, mark schemes and Principal Examiner Report supplied by the user. All practice questions are newly written.